Problem statement: Create a classification model to predict the sentiment either (1 or 0) based on Amazon Alexa reviews.

Tools Used:

Python
scikit-learn
Nlp

CODE:

Importing Required Libraries

In [1]:

```
import numpy as np
import pandas as pd
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense,Dropout,LSTM,Embedding
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from sklearn.model_selection import train_test_split
from nltk.corpus import stopwords
import re
import matplotlib.pyplot as plt
```

In [2]:

```
data = pd.read_csv("/kaggle/input/amazon-alexa-reviews/amazon_alexa.tsv",sep='\t')
```

In [3]:

```
data.head()
```

Out[3]:

	rating	date	variation	verified_reviews	feedback
0	5	31-Jul-18	Charcoal Fabric	Love my Echo!	1
1	5	31-Jul-18	Charcoal Fabric	Loved it!	1
2	4	31-Jul-18	Walnut Finish	Sometimes while playing a game, you can answer...	1
3	5	31-Jul-18	Charcoal Fabric	I have had a lot of fun with this thing. My 4 ...	1
4	5	31-Jul-18	Charcoal Fabric	Music	1

In [4]:

```
data['rating'] = data['rating'].apply(lambda x: 1 if x >= 3 else 0)
data = data[['rating','verified_reviews']]
```

In [5]:

```
data.head()
```

Out[5]:

	rating	verified_reviews
0	1	Love my Echo!
1	1	Loved it!
2	1	Sometimes while playing a game, you can answer...
3	1	I have had a lot of fun with this thing. My 4 ...
4	1	Music

In [6]:

```
data['rating'].value_counts()
```

Out[6]:

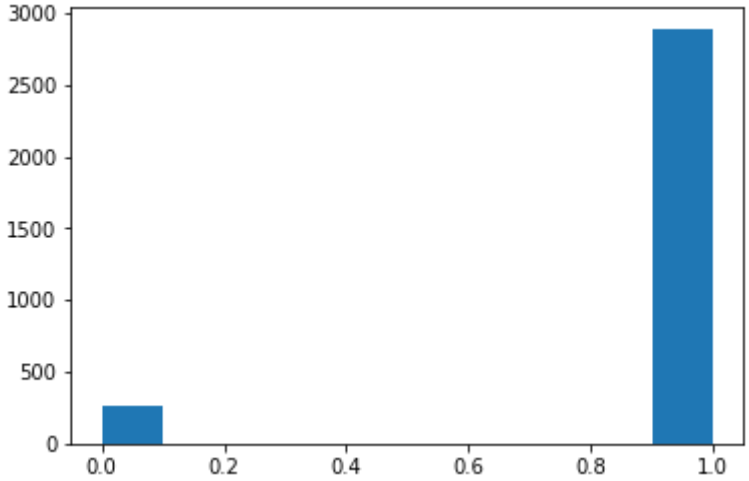
```
1    2893
0     257
Name: rating, dtype: int64
```

In [7]:

```
plt.hist(data['rating'])
```

Out[7]:

```
(array([ 257.,    0.,    0.,    0.,    0.,    0.,    0.,    0.,    0.,
        2893.]),
 array([0. , 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1. ]),
 <BarContainer object of 10 artists>)
```



Time to Clean data

In [8]:

```
STOPWORDS = stopwords.words("english")
```

In [9]:

```
def clean(x):
    x = x.lower()
    x = re.sub("[^\w\d]"," ",x)
    x = " ".join([y for y in x.split() if y not in STOPWORDS])
    return x
```

In [10]:

```
data['text'] =  data['verified_reviews'].apply(lambda x :  clean(x))
```

In [11]:

```
data.head()
```

Out[11]:

	rating	verified_reviews	text
0	1	Love my Echo!	love echo

	rating	verified_reviews	text
1	1	Loved it!	loved
2	1	Sometimes while playing a game, you can answer...	sometimes playing game answer question correct...
3	1	I have had a lot of fun with this thing. My 4 ...	lot fun thing 4 yr old learns dinosaurs contro...
4	1	Music	music

In [12]:

```
np.array(data['text'][0:5])
```

Out[12]:

```
array(['love echo', 'loved',
       'sometimes playing game answer question correctly alexa says got wrong
answers like able turn lights away home',
       'lot fun thing 4 yr old learns dinosaurs control lights play games like
categories nice sound playing music well',
       'music'], dtype=object)
```

Cleaning works well !

Splitting the data into train/test

In [13]:

```
x=  np.array(data['text'])
y = np.array(data['rating'])
```

In [14]:

```
x[:3]
```

Out[14]:

```
array(['love echo', 'loved',
       'sometimes playing game answer question correctly alexa says got wrong
answers like able turn lights away home'],
      dtype=object)
```

In [15]:

```
y[:3]
```

Out[15]:

```
array([1, 1, 1])
```

In [16]:

```
data['rating'].shape
```

Out[16]:

```
(3150,)
```

In [17]:

```
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.1,shuffle=True,ra
ndom_state=42,stratify=y)
```

In [18]:

```
x_train.shape
```

Out[18]:

```
(2835,)
```

Time convert text data to binary format

In [19]:

```
token = Tokenizer(num_words=5000)

token.fit_on_texts(x_train)
x_train = token.texts_to_sequences(x_train)
x_test = token.texts_to_sequences(x_test)
```

In [20]:

```
##### Hyperparamaters
vocab_size = len(token.word_index) +  1
maxlen = 30
```

In [21]:

```
x_train[:5]
```

Out[21]:

```
[[96],
 [260, 2, 39, 1, 74, 63, 109, 290, 146],
 [3, 16, 6, 57, 49, 64],
 [567, 1, 205, 50, 78, 3],
 [4, 206, 851, 941, 63, 10]]
```

In [22]:

```
x_train = pad_sequences(x_train,padding='pre',maxlen=maxlen)
x_test = pad_sequences(x_test,padding='pre',maxlen=maxlen)
```

In [23]:

```
x_train
```

Out[23]:

```
array([[   0,    0,    0, ...,    0,    0,   96],
       [   0,    0,    0, ...,  109,  290,  146],
       [   0,    0,    0, ...,   57,   49,   64],
       ...,
       [   0,    0,    0, ...,  321,  268,  105],
       [   0,  352, 1583, ...,    5,  121,  692],
       [   0,    0,    0, ...,  337, 1853, 1854]], dtype=int32)
```

Time to Create a neural network

In [24]:

```
model = Sequential([
    Embedding(vocab_size,100,input_length=maxlen),
    Dropout(0.2),
    LSTM(200),
    Dropout(0.2),
    Dense(1,activation='sigmoid')
```

```
])
```

```
2022-09-21 06:36:04.289226: I tensorflow/core/common_runtime/process_util.cc:1
46] Creating new thread pool with default inter op setting: 2. Tune using inte
r_op_parallelism_threads for best performance.
```

In [25]:

```
model.compile(loss='binary_crossentropy',optimizer='adam',metrics=['acc'])
```

In [26]:

```
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 30, 100)	382600
dropout (Dropout)	(None, 30, 100)	0
lstm (LSTM)	(None, 200)	240800
dropout_1 (Dropout)	(None, 200)	0
dense (Dense)	(None, 1)	201

Total params: 623,601
Trainable params: 623,601
Non-trainable params: 0

In [27]:

```
history  = model.fit(x_train,y_train,epochs=10,batch_size=32,validation_data=(x_te
st,y_test))
```

```
Epoch 1/10
2022-09-21 06:36:04.884279: I tensorflow/compiler/mlir/mlir_graph_optimization
_pass.cc:185] None of the MLIR Optimization Passes are enabled (registered 2)
89/89 [==============================] - 9s 65ms/step - loss: 0.2978 - acc: 0.
9139 - val_loss: 0.2460 - val_acc: 0.9175
Epoch 2/10
89/89 [==============================] - 6s 69ms/step - loss: 0.1738 - acc: 0.
9418 - val_loss: 0.1754 - val_acc: 0.9270
Epoch 3/10
89/89 [==============================] - 5s 58ms/step - loss: 0.0837 - acc: 0.
9686 - val_loss: 0.1302 - val_acc: 0.9429
Epoch 4/10
89/89 [==============================] - 5s 60ms/step - loss: 0.0509 - acc: 0.
9831 - val_loss: 0.1725 - val_acc: 0.9365
Epoch 5/10
89/89 [==============================] - 5s 60ms/step - loss: 0.0317 - acc: 0.
9898 - val_loss: 0.1617 - val_acc: 0.9524
Epoch 6/10
89/89 [==============================] - 5s 59ms/step - loss: 0.0330 - acc: 0.
9915 - val_loss: 0.1947 - val_acc: 0.9429
Epoch 7/10
89/89 [==============================] - 6s 68ms/step - loss: 0.0231 - acc: 0.
9937 - val_loss: 0.3817 - val_acc: 0.9429
Epoch 8/10
```

```
89/89 [==============================] - 5s 60ms/step - loss: 0.0249 - acc: 0.
9922 - val_loss: 0.2647 - val_acc: 0.9397
Epoch 9/10
89/89 [==============================] - 5s 60ms/step - loss: 0.0275 - acc: 0.
9922 - val_loss: 0.1707 - val_acc: 0.9397
Epoch 10/10
89/89 [==============================] - 5s 60ms/step - loss: 0.0275 - acc: 0.
9908 - val_loss: 0.2882 - val_acc: 0.9397
```

In [28]:

```
loss,acc =  model.evaluate(x_test,y_test)
acc
```

```
10/10 [==============================] - 0s 15ms/step - loss: 0.2882 - acc: 0.
9397
```

Out[28]:

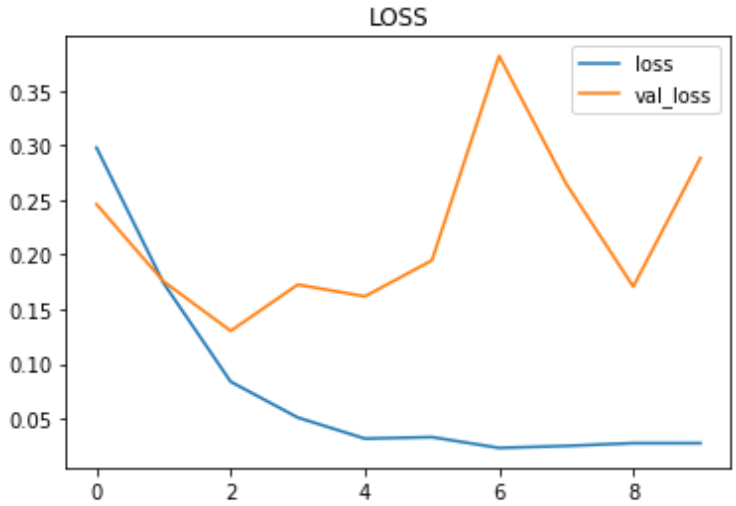
```
0.9396825432777405
```

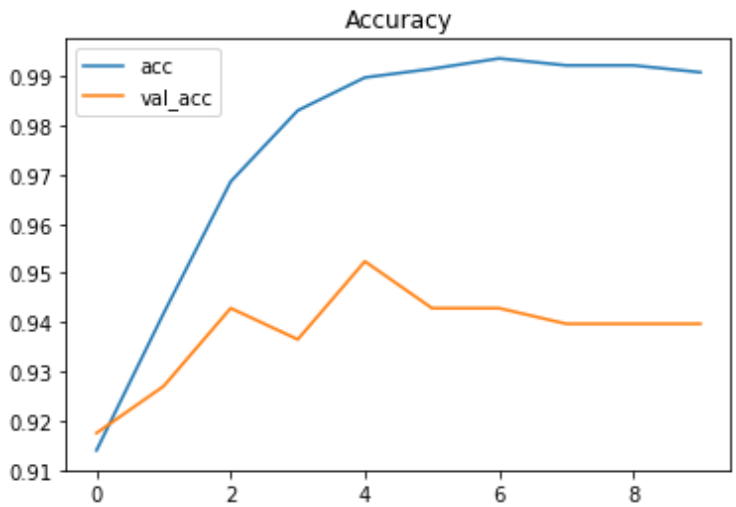
In [29]:

```
history_df = pd.DataFrame(history.history)
history_df[['loss','val_loss']].plot(title='LOSS')
history_df[['acc','val_acc']].plot(title='Accuracy')
```

Out[29]:

```
<AxesSubplot:title={'center':'Accuracy'}>
```





NOTE : I have also testted the model on convolution model and it gives acc of about 93%.

MAKING PREDICTIONS

In [30]:

```
def preprocess(text):
    text = token.texts_to_sequences(text)
    text =  pad_sequences(text,padding='pre',maxlen=maxlen)
    return text
```

In [31]:

```
def predict(text):
    text = preprocess(text)
    pred = model.predict(text)
    return pred
```

In [32]:

```
predict(["like siri fact siri answers accurately alexa see real need household tho
ugh good bargain prime day deals"])
```

Out[32]:

```
array([[0.00350943]], dtype=float32)
```